Verilog HDL

HDL gives output as logic circuit

Assignments

Assignments are used to assign value into nets and variables.

Three types:

* Continuous, procedural, produral continuous

Continuous assignment

> Assign a value to net (wire) continuously, whenever the input to the assignment statement change

Ex: assign c=a+b

> It only used in combinational logic circuit

RTL components

> Combinational logic

> Sequential logic

> Registers

> Data + control path

> FSMs

> Timing : setup/hold, clocking

> Reset strategies (sync/async)

Digital logic fundamentals

> Logic gate are the fundamental building blocks of digital electronics.
> They take one or more binary inputs (0 and 1) to produce single binary output.

Basic logic gates

- AND Gate → $A . B = Y$
  Output 1 if both inputs are 1.
- OR Gate → $A + B = Y$
  Output 1 if atleast one input is 1.
- NOT Gate → $Y = \bar{A}$
  It takes single input and reverses it.

Universal gate

- NAND Gate → $Y = \overline{A . B}$
  This is an AND gate followed by NOT gate
- NOR Gate → $Y = \overline{A + B}$
  This is an OR gate followed by NOT gate

The Special gate

- XOR gate → $Y = A \oplus B$
  Output 1 if the inputs are different

Simulation results

By observing the simulation results, when the simulation waveform is formed, It shows clock pulse wave as 0 and 1 uniformly. and reset time wave also formed.

https://www.edaplayground.com/x/CFYu

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